

SIGIL CR

Kerala, India (Open to Relocate to Germany)

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Summary

HVAC Design Engineer with 8 years of experience delivering end to end HVAC system design across commercial institutional and mixed use projects. Strong expertise in heat load calculation ventilation design BIM coordination and construction documentation. Extensive knowledge of ASHRAE, CIBSE, SMACNA standards and green building compliance frameworks. Proven capability in client coordination equipment selection and zero clash Revit model delivery. Drives energy efficient and technically compliant HVAC solutions.

Core Skills

- **HVAC System Design:** Cooling and Heat Load Calculation (TRACE 700, HAP), Ventilation Design, Duct Sizing, Equipment Selection, Plant Layout Planning, Chiller Piping, Pump Room Design
- **Standards and Compliance:** ASHRAE, CIBSE, SMACNA, DW144, DW172, Green Building Regulations
- **Project Delivery and Coordination:** Client Coordination, RFI Management, Team Supervision, Quality Control
- **BIM and Modeling:** Autodesk Revit, AutoCAD 2D and 3D, Navisworks Clash Detection, Bluebeam Coordination

Experience

Vee Technologies Pvt Ltd

March 2022 - Present

Senior Design Engineer - I

Karnataka, India

- Lead end to end HVAC design execution for 10 large scale projects, conducting client meetings and translating technical requirements into coordinated design solutions.
- Direct 5 member design team, allocate work packages, perform structured QC reviews, and maintain project tracking logs to ensure schedule adherence.
- Conduct Bluebeam coordination sessions with multidisciplinary stakeholders, resolving RFIs and reducing design conflicts by 75%.
- Execute heat load calculations using TRACE 700 parallel to Revit modeling to optimize equipment sizing and ventilation performance.

Archingen India Pvt Ltd

August 2019 - February 2022

Mechanical Designer - I

Kerala, India

- Designed ventilation systems and duct layouts in Revit and AutoCAD across 15 commercial and mixed use projects.
- Developed water supply, drainage, storm water, chiller piping, and pump room layouts aligned to project specifications.
- Prepared coordinated construction documents and supported multidisciplinary service integration.
- Performed equipment selection and ventilation studies ensuring compliance with ASHRAE and SMACNA standards.

Cadeosys Engineering Consultant

June 2018 - August 2019

Mechanical Engineer

Kerala, India

- Delivered ventilation system design and drafting using Revit and AutoCAD for commercial developments.
- Prepared shop drawings and mechanical service layouts supporting site level execution.
- Assisted in equipment sizing and HVAC calculations aligned to international standards.
- Supported inter discipline coordination and drawing revisions improving approval timelines by 15%.

Engineering Consortium Consulting Engineers

February 2018 - June 2018

Trainee Mechanical Design Engineer

UAE

- Performed cooling load calculations using HAP and prepared detailed thermal load summaries.
- Developed ventilation calculations in accordance with ASHRAE, CIBSE, SMACNA, DW144, and DW172 standards.
- Supported green building compliance documentation aligned to Dubai Municipality regulations.
- Assisted senior engineers in BIM model updates and construction documentation preparation.

Jamal Cooperation

June 2015 - January 2018

Project Engineer

Kerala, India

- Executed manual heat load calculations using ASHRAE pocket guide methodologies for 12 building projects.
- Prepared BOQ documentation and shop drawings supporting cost estimation and procurement.
- Supervised HVAC unit installation and duct fabrication ensuring adherence to design specifications.
- Coordinated on site mechanical activities improving installation accuracy and reducing rework by 18%.

Education

Anna University

2011 - 2015

Bachelor of Engineering - Mechanical Engineering

India

Languages

- English - Full Professional Proficiency (C2)
- German - Intermediate Proficiency (B1)